Paleomagnetic-Tectonic Study of Mesozoic Dike Swarms in the Appalachians

JELLE DE BOER

Wesleyan University, Middletown, Connecticut

Paleomagnetic evidence indicates that most of the extensive dike swarms cutting Triassic and older formations probably intruded in a time of regional tectonic and magmatic activity distinct from the late Triassic tectogenesis. The fossil magnetic directions of the dikes coincide neither with Late Triassic nor with Early Cretaceous paleomagnetic directions, which suggests a Jurassic age for the intrusions. The dikes were emplaced along tensional fractures that were the surficial expressions of deep-seated movements. The fan-shaped arrangement of these fractures indicates a southwestward decrease of the rotational component of the shear couple and a sinistral polarity of the shear movements. Periodic increases in tectonic and mafic magmatic activity, such as this one in the Jurassic, appear to be characteristic of Mesozoic deformations in the Appalachians.

INTRODUCTION

The latest tectonic map of the United States [*Cohee et al.*, 1962] only recently drew general attention to the extensive dike swarms cutting Triassic and older formations in the Appalachians, a topic heretofore almost neglected by geologists. Interest in these intrusions undoubtedly was stimulated by *King*'s [1961] paper, which pointed out their widespread occurrence. These Mesozoic dikes give evidence on what is, presumably, one of the latest periods of regional igneous activity in the eastern United States. They are generally referred to as Late Triassic dikes and have been correlated recurrently with the basaltic sills and flows intercalated in the Late Triassic Newark formation.

DIKE PROPERTIES

The dikes extend throughout the Appalachians from Maine to Alabama. They may be present in southern Alabama, covered by sedimentary deposits of the Atlantic coastal plain. Only a few dikes of the 'Triassic type' are known in Maine, and even fewer have been reported from the Canadian Appalachians. Their eastward extent into the area covered by the Cretaceous and younger coastal-plain deposits is unknown. To the west, very few of the Piedmont dikes are found beyond the Brevard fault [*Lester and Allen*, 1950]. In the central Appalachians relatively few Mesozoic dikes have intruded the formations of the Blue Ridge anticlinorium and the Valley and Ridge province.

Data on the dikes and their pattern are as follows:

1. The Mesozoic dike swarms are concentrated in a relatively narrow zone extending NE-SW along the main Appalachian axis. The presently known dike distribution appears to indicate a significant decrease in dike frequency from the Piedmont northeastward into New England. Numerous dikes have been mapped, however, in the Boston area and in Maine. The age of these dikes is for the most part unknown; therefore, it is possible that the anomalously low dike frequency is confined to southern New England and is a local phenomenon only.

2. The dike trends are dominantly discordant to those of the older structures. In the southwestern segment of the Appalachians the dikes extend almost at right angles to the general structures: in the central and northeastern segments they cut the older structures diagonally (see Figure 1).

In the Piedmont the dikes trend consistently NNW, at angles ranging from 65° to 80° to the regional trend of the older structures. The consistency in dike trend over large distances is remarkable, particularly since the dikes cut metamorphic formations of widely different textures and structures.

There are some complexities of trend in the central segment of the Appalachians, and in

Fig. 1. Distribution of Mesozoic dikes in the Appalachians.

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Virginia and Pennsylvania the dike trends show maxima in three different directions. In northwestern Virginia the dikes trend NW, N, and NE, in southeastern Virginia they trend NNW, N, and NNE, and in southeastern Pennsylvania they trend NNE, NE, and ENE. These distributions appear to suggest that the magma intruded one set of tensional and two sets of related shear faults. The large differences in apparent conjugate shear angles are, however, highly suspect and seem to imply that some of the magma followed older fractures.

In northwestern Virginia the regional trend of the older structures is about NNE. One set of dikes accurately follows this trend, which is especially evident in the southern part of the Culpeper basin. The NW-trending set is at right angles to the regional trend and is best developed in the folded belt, where it parallels the direction of the theoretical cross faults expected in such a folded rock sequence. In southeastern Pennsylvania the ENE-trending dikes closely follow the regional structural trend. If only the dike concentrations that cut the Newark formation in the Late Triassic rift zones are considered, it is the central set that dominates in all three frequency diagrams.

In the New England segment of the Appalachians, the dike trend is consistently NNE, and a concentration similar to the one in the Piedmont segment has been found. The angle between the dominant dike trends and the regional structure is about 20°.

3. In general, the dikes are steeply inclined. There appears to be no preferred dip direction: measurements on 87 dikes show that the dips are distributed more or less symmetrically around the vertical. Locally, as in the southern Piedmont, the dip is invariably eastward. The dikes that cut the Late Triassic Newark series do not show such local asymmetries, which suggests that they intruded after the major tilting of the series. This would mean that considerable time may have elapsed between the emplacement of these dikes and the intrusion of the Late Triassic sills and lava flows intercalated in the Newark beds.

4. The dikes are remarkably straight and extend over long distances. Their surface outcrops are very discontinuous laterally. Magnetometer surveys indicate that some of the dikes simply continue at depth. The vertical sinuosities locally coincide with variations in the thickness of a dike and thus can be related to lateral differences in the width of the original fracture zones. Several of the dikes can be traced along strike for more than 150 km. Their great length is a measure of their regional importance and great vertical extent.

5. The dikes are relatively narrow; their widths vary from a few to a hundred meters, only rarely exceeding a hundred meters. No evidence has been found of composite dikes resulting from repeated filling of the fractures. The absence of this type of dike indicates a one-phase origin and also suggests that the extensive fractures followed by the dikes opened up in a relatively short time.

6. The dike rock is basaltic, monotonously uniform in composition. In addition to its normal augite-labradorite assemblage, the most common accessory minerals present are olivine, quartz, magnetite, and ilmenite. The magnetite occurs in a variety of forms, most frequently as finely dispersed subhedral to anhedral grains in the groundmass. It also appears as single euhedral skeletal crystals, irregular intergrowths in the pyroxenes, scattered grains bordering the pigeonite, interstitial grains along the cleavage planes in hypersthene, and local concentrations enclosed in the deuteric minerals. Lamellar intergrowths of leucoxene and magnetite, indicating altered titaniferous magnetites, are commonly observed.

7. The dike swarms are generally referred to as Late Triassic intrusions. Repeatedly they have been correlated with the basaltic sills and flows intercalated in the Late Triassic Newark formation. Because the dikes cut the youngest sediments of the Newark formation, however, they cannot be so correlated, and the only feature the dikes and flows have in common is their composition.

Randazzo and Wheeler [1965] reported a typical Triassic dike in the Wadesboro region of North Carolina that intruded the unconsolidated post-Newark sediments of the coastal plain. According to *Lester and Allen* [1950] none of the Piedmont dikes could be traced to a contact with the sediments overlying the Newark series. *Reinemund* [1955] reported that the dikes of the Deep River basin pass beneath the sand and gravel deposits of the coastal plain, which may be of Cretaceous or Tertiary age.

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Thus, the time during which the dikes could have intruded ranges from late Late Triassic to Early or Middle Cretaceous.

Tectonic Activity before and after the Dike Emplacement

Little is known about Mesozoic tectonic and magmatic activity in the Appalachians. The Late Triassic tectogenesis has been studied by *Reinemund* [1955] in the Deep River basin and by *Sanders* [1963] in the Connecticut Valley. Isotopic age determinations of separate intrusions in the volcano-plutonic complexes of the White Mountains (New Hampshire) and the Monteregian Hills (southeastern Quebec) indicate a continuation of igneous and tectonic activity into Early Cretaceous time. From these rather isolated pieces of information the following sequence of events can be derived:

1. An initial period of graben subsidence, mostly along lines of weakness trending NE–SW; syntectonic deposition of Late Triassic clastic sediments in a continental environment; contemporaneous mafic igneous activity resulting in the emplacement of extensive sills and the outpouring of extensive flows.

The agreement of the directions of remanent magnetization in the sediments and in intrusive igneous rocks of New Jersey and Pennsylvania [*Opdyke,* 1961] indicates that most of the sedimentary beds were nearly horizontal at the time of igneous activity. The age of the Palisades sill has been established at 190 m.y. [*Erickson and Kulp,* 1960].

2. A phase of longitudinal crustal arching in the center of the original graben, more or less parallel to the central Appalachian axis, resulting in general uplift and differential tilting of the Newark series.

Beck's [1965] paleomagnetic study on the Birdsboro and Gettysburg basaltic intrusions suggests that the deformation of these masses can be explained by two subsequent processes: a differential rotation of 20° to 30° around their local strike line and a superimposed general northwestward uplift. The differential rotation may be due to drag phenomena along the northern border fault of the Gettysburg-Newark graben.

3. A second period of graben subsidence, accompanied by the formation of longitudinal folds for which no record of syntectonic sedimentation has been preserved.

4. A period of increased tectonic activity, resulting in strike-slip displacements along older lineaments and the subsequent formation of transverse or cross folds. The final stage of this period may have been characterized by the development of extensive fractures, subsequently injected by mafic dikes. These dikes are the subject of this paper. *Sanders* [1963] was the first to point out the significance of two periods of igneous activity separated by a period of deformation in early Mesozoic time.

5. A period of local tectonic and magmatic activity as indicated by the intrusions of the Ascutney gabbro in the White Mountains complex and the Monteregian and Megantic gabbros in southeastern Quebec. The Mount Ascutney gabbro was dated at 125–135 m.y. [*Faul et al.,* 1963]; the Monteregian gabbros, at 124 m.y.; and the Mount Megantic gabbro, at 113 m.y. [*Larochelle,* 1961, 1962].

Whether the dikes were emplaced during the Late Triassic period of tectogenesis or intruded in a separate period of structural and magmatic activity will remain a problem as long as no absolute age determinations are available. Paleomagnetic data may help to solve this problem, one of great significance in deciphering the structural evolution of the Appalachians during Mesozoic time.

Paleomagnetic Data

A paleomagnetic study was made by the author to obtain more detailed information on the age and relationships of the dikes. The fossil directions of magnetization in the dikes were then correlated with previously determined magnetic directions.

Available Triassic paleomagnetic poles are listed in Table 1. The most extensive work was done by *Bowker* [1960], who collected about 600 cores from 77 outcrops throughout the Appalachians from Nova Scotia to North Carolina. *Opdyke* [1961] and *Beck* [1965] obtained very reliable data in New Jersey and Pennsylvania. A combination of their results and the results of Bowker makes it possible to apply a certain correction to the older data published by *Du Bois et al.* [1957]. The latitude of the Late Triassic magnetic pole appears to have been close to 62°N (which is approximately

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TABLE 1. Paleomagnetic Data for the Triassic and Cretaceous Periods in the Eastern United States

Sampling Location	Cores	Specimens	Rock Type*	α_{95}†	K‡	Magnetic Pole Position: Latitude	Magnetic Pole Position: Longitude	Reference
				Triassic				
Nova Scotia	4	11	I	4.5	137	70.5°N	71.5°E	*Bowker* [1960]
	8	24	I	28.5	4.7	74°N	98.5°E	
	9	32	I	20.5	7.1	59.5°N	153.5°E	
	2	12	I	35	54	70°N	148°E	
	6	38	I	35.5	4.5	74.5°N	13.5°W	
	5	16	I	16	29	77.5°N	119.5°E	
	7	14	I	9.5	42	71°N	26°E	
	7	17	I	3	345	68.5°N	40.5°E	
	7	15	I	8	59	67°N	22.5°E	
	7	12	I	12.5	24	75.5°N	174.5°E	
	8	15	I	2	667	61.5°N	51.5°E	
	74	217	I	11.5	14	77°N	75.5°E	*Cox and Doell* [1960]
Massachusetts	8	16	I	5.5	105	43°N	50.5°E	*Bowker* [1960]
	4	16	I	27	13	58.5°N	7°E	
	8	13	I	11.5	24	46°N	48.5°E	
	3	6	I	8	125			
	3	6	I	5	250			
	3	6	I	8	100	55°N	88°E	*Irving and Banks* [1961]
	4	8	I	9	82			
	3	6	I	17	23			
	8	8	I	15	...	54°N	90°E	*Du Bois et al.* [1957]
Connecticut	12	32	M	15	7	53°N	86°E	
	8	12	I	6.5	72	58°N	44°E	*Bowker* [1960]
	6	8	I	15	22	75°N	13°W	
	6	8	I	15	22	75°N	13°W	
	3	5	I	5	625	52.5°N	77.5°E	
New Jersey	7	7	S	4.5	168	63.5°N	107.5°E	*Bowker* [1960]
	6	9	S	7.5	43	65°N	66°E	
	11	13	S	10	36	73.5°N	81.5°E	
	8	11	S	16.5	18	61.5°N	139.5°E	
	48	211	S		46			
	17	69	I		38	63°N	108°E	*Opdyke* [1961]
	13	50	I		100			
Pennsylvania	8	12	I	4	176	68°N	93°E	*Bowker* [1960]
	6	7	I	3.5	394	62.5°N	120°E	
	8	8	I	8	48	66.5°N	131.5°E	
	9	15	I	2.5	396	63.5°N	117°E	
	8	10	I	3	389	71.5°N	136.5°E	
	8	19	I	3	327	62.5°N	102°E	
	6	6	I	10	48	52.5°N	151°E	
	6	10	S	6.5	69	69°N	105.5°E	
	9	11	S	7.5	46	73.5°N	122°E	
	10	14	S	6.5	58	61.5°N	113.5°E	
	33	33	I					
	23	23	I	3.3	102	62°N	105°E	*Beck* [1965]
	39	39	I					
Maryland	9	12	S	6	71	62.5°N	104°E	*Bowker* [1960]
	15	20	S	3	161	61.5°N	91°E	
	6	6	S	19.6	13	52.5°N	115°E	

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TABLE 1. (Continued)

Sampling Location	Cores	Specimens	Rock Type*	α_{95}†	K‡	Magnetic Pole Position: Latitude	Magnetic Pole Position: Longitude	Reference
Virginia	15	24	I	4.5	77	62°N	125°E	*Bowker* [1960]
	8	14	I	11	26	58.5°N	172.5°E	
	8	16	I	12.5	20	88°N	6.5°W	
	6	7	S	4.5	202	59°N	125°E	
	8	14	S	15	15	25°S	118°W	
	7	7	S	7	81	70.5°N	91.5°E	
	9	12	S	21.5	6.7	67°N	7°E	
	11	16	S	6	63	62°N	74.5°E	
	10	12	S	9.5	28	70°N	134.5°E	
Carolina	8	14	S	25	5.9	38°S	77.5°W	*Bowker* [1960]
	8	12	S	14	16	53.5°N	73°E	
	12	24	S	15	9.4	23°S	94.5°W	
	8	9	S	3	330	58°N	97.5°E	
	9	18	S	19.5	3.3	33.5°S	81°W	
	7	9	S	10.5	34	59.5°N	92.5°E	
				Cretaceous				
Quebec (Canada)	12	12	I	6.5	44	68.5°N	171.5°E	*Larochelle* [1962]
	10	10	I	18	7.9			
	39	39	I	9	7.5	65°N	157°W	
Ringness Island (Canada)	20	40	I	8	20	69°N	180°	*Larochelle and Black* [1963]
California	181	181	I					
	5	5	M	9.6	18	69°N	165°W	*Grommé and Merrill* [1965]
	17	17	M					
Arizona	6	10	S	7		74°N	158°W	*Runcorn* [1956]

* I: igneous; S: sediments; M: mixed rocks.
† Radius of cone of confidence at 95% probability level.
‡ Precision parameters.

10° farther north than had been originally indicated), and the longitude seems to have been near 100°E. *Beck,* 1965, estimated that the average geomagnetic pole for the North American Triassic was located at 62°N, 101°E in present-day central Siberia.

Almost no Jurassic poles are known. *Collinson and Runcorn* [1960] listed those for the Kayenta and Carmel formations in Arizona and Utah; *Cox and Doell* [1960] calculated average poles from their data: 83°N, 63°E for the Kayenta formation and 80°N, 160°W for the Carmel formation. Both average poles are derived from magnetic directions that cluster around the present magnetic pole, suggesting the possibility of magnetic instability.

Few poles are known for the Cretaceous—these are, however, well dated and reliable (see Table 1). The poles for the eastern United States were obtained by *Larochelle* [1962] from the Monteregian Hills in southeastern Quebec. This linear group of mafic intrusions has recently been dated as Early Cretaceous (124 m.y. Mount Megantic, a single intrusion east of the Monteregian Hills, seems to belong to the same igneous complex; it was emplaced about 113 m.y. ago. The paleomagnetic poles obtained from granitic plutons in California by

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Grommé and Merrill [1965] are also listed in Table 1. The agreement among all these Cretaceous determinations is surprising. The average geomagnetic pole for the North American Cretaceous appears to have been at 69°N, 165°W in the present-day Bering Straits.

Thus, the Mesozoic polar path on the North American continent appears to have been almost latitudinal. From a well verified Late Triassic location near 62°N, 101°E the magnetic pole may have moved initially northward to latitude 80°N and from there slipped east-southeastward through about 90° of longitude toward an early Cretaceous position at 69°N, 165°W (see Figure 3). More detailed information about this polar path, currently being acquired, is important because polar shift is useful in correlation, if used in conjunction with magnetic pole determinations from rock units or formations of known age.

In the present study, 80 dikes in a region extending from Connecticut to Georgia were sampled. Cores were taken with a portable diamond drill and were oriented in the field with a Brunton compass. At least one core was taken from each dike after making an examination of the exposure for possible lightning effects and tectonic disturbances. Wherever exposures permitted, two cores were taken—one from the chilled border and one from the coarse central part of the dike—in order to average out any differences in magnetic properties caused by variations in the cooling rate. Most cores yielded two or more specimens, but the uppermost one generally could not be used because of its visibly weathered zone, in no case more than a quarter of an inch deep, since the freshest part of each exposure was chosen for coring.

The specimens were measured with a spinner magnetometer designed by *Doell and Cox* [1965]. The estimated accuracy is greater than 2°. A plot of magnetic directions obtained from the first set of measurements showed the usual large scatter. The data showed also that the sampled exposures of 7 of the dikes (9 cores) had been displaced before the coring, primarily by the sliding of blocks during weathering. In addition, about 15 cores from other dikes had to be discarded for the same reason. The rejected magnetic directions fell outside a circle of 50° radius with its center within the largest concentration of directions. These deviating directions continued to be different from the others during demagnetization. To reduce the initially large scatter, the samples were subjected to an alternating-field demagnetization in steps of 25 oe. Rocks of similar lithology in the sampled area have been considered magnetically 'clean' after alternating-field demagnetization in peak fields of 150 oe [*Opdyke*, 1961; *Beck*, 1965]. However, a plot of the magnetic directions measured after the rocks had been subjected to a magnetic field of this intensity showed that the scatter had increased. It was decided to continue the demagnetization in steps of 50 oe up to 700 oe. Plots of the magnetic directions measured after subjecting the samples to 200, 300, and 400 oe also showed no significant decrease of scatter.

The scatter could be reduced considerably by considering each core separately, and by using the criterion that the direction of the stable component can be obtained after the vector ceases to rotate (only intensity changes occur). The peak fields in which the samples could be considered clean ranged from 100 to 450 oe. Most magnetic directions that had been more or less constant over a certain interval started to deviate again when subjected to fields of 500 oe or higher. No definite peak at 150 oe appears in a histogram showing the number of samples plotted versus the minimum field-intensity value at which the viscous remanent component is considered eliminated. Instead, there is an irregular curve with various maxima between 150 and 400 oe (see Figure 2). Therefore it seems highly necessary to consider each sample separately during demagnetization.

No corrections were applied for post-emplacement movement of the dikes. They show the same subvertical dip whether intruding eastward- or westward-dipping Triassic sediments,

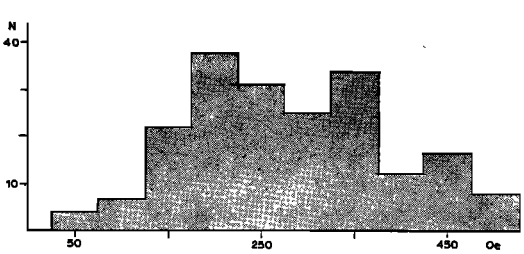


Fig. 2. Histogram of peak-demagnetization values (in oersteds).

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which indicates intrusion after the main tilting of the Newark formation. The fact that there are no systematic differences in the declination of the magnetic directions for the various Appalachian localities shows that the fanlike arrangement of the dikes is not due to differential rotation.

Paleomagnetic data on the dikes are listed in Table 2. The declinations of the magnetic directions in the dikes vary from about N–S to NW–SE. There is no indication of a relationship between magnetic declination and dike trend. *Strangway's* [1961] study of Precambrian diabase dikes in the Canadian shield indicated that the average declinations of the remanent magnetic directions were located within 24° and, in most cases, within 15° of the strike directions of the dikes. Thus dikes with widely differing strike directions showed a stable component of magnetization more or less parallel to the plane of intrusion. Since most workers on paleomagnetism assume that the magnetic directions acquired and retained by a rock are parallel to the direction of the earth's magnetic field at the time the rock became magnetized, Strangway's hypothesis would mean that remanent magnetic directions of dikes cannot be used for age correlations.

Strangway explained his results on the basis of self-demagnetization within an infinite sheet. This requires that a small portion of the magnetic minerals have a lower Curie temperature and a greater stability than the bulk of the rock's magnetic minerals. The minerals with the higher Curie temperature produced an internal demagnetization during the cooling, which tended to cancel out the component of the earth's magnetic field normal to the dikes. The minerals with the lower Curie temperature then acquired a magnetic moment parallel to the internal field of the dike, which would be nearly in the plane of the dike. *Fahrig et al.* [1965], in a study of diabase dikes of the Canadian shield, pointed out that, if the magnetization directions of a dike were completely random, 40% of all mean vectors would lie within 24° of any chosen plane. Therefore, there is a 40% chance that the mean vector of any dike swarm will lie within 24° of the plane of the dike swarm. Fahrig and his coworkers concluded from their study, which included dikes examined by Strangway, that there is lack of evidence for a relationship between direction of magnetization and orientation of the plane of the dike. Our magnetic data on the Appalachian dikes confirm this conclu-

TABLE 2. Paleomagnetic Data for the Mesozoic Dikes in the Appalachians

Sampling Location	Nd*	Cores	Specimens	D†	I†	K‡	α_{95}§	VGP¶	
Georgia									
Piedmont (33.5°N, 83.5°W)	10	14	21	340	+25	37	8	63N	142E
North South Carolina									
Deep River basin (35.5°N, 79.5°W)	18	26	48	342	+31	26	7	65N	145E
North Carolina									
Danville basin (36.5°N, 79.5°W)	6	8	15	333	+32	31	12	59N	159E
Virginia									
Richmond basin (37.5°N, 77.5°W)	3	6	9	345	+35	43	19	67N	142E
Bent Creek Warren basin Nelson County Farmville basin (37.5°N, 78.5°W)	15	22	37	340	+36	44	6	65N	152E
Augusta County (38°N, 79°W)	6	7	13	347	+37	65	8	70N	137E
Pennsylvania									
Gettysburg basin (40°N, 77°W)	6	12	24	352	+40	47	10	72N	127E
Newark basin (40.5°N, 75°W)	5	10	17	346	+41	61	10	69N	143E
Connecticut									
Whole state (41.5°N, 72°W)	5	16	39	343	+38	107	7	65N	148E

* Nd: number of sampled dikes.
† Declination and inclination of average magnetic vector.
‡ Precision parameter.
§ Radius cone of confidence at 95% probability level.
¶ Virtual Geomagnetic pole position.

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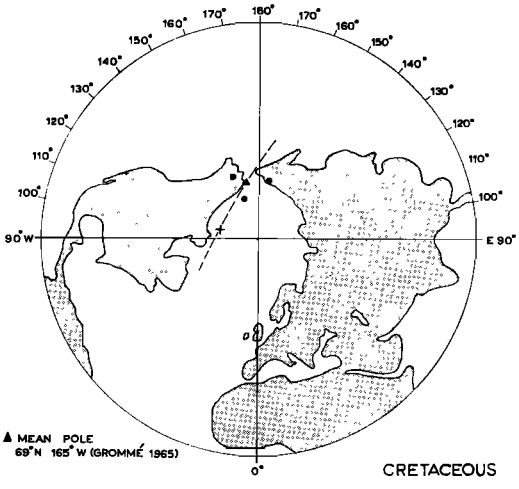


Fig. 3a.

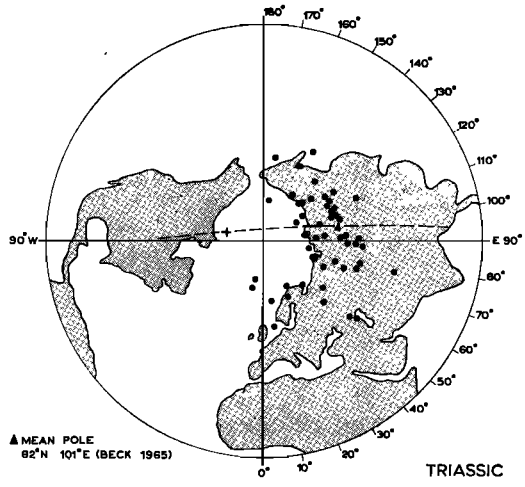


Fig. 3c.

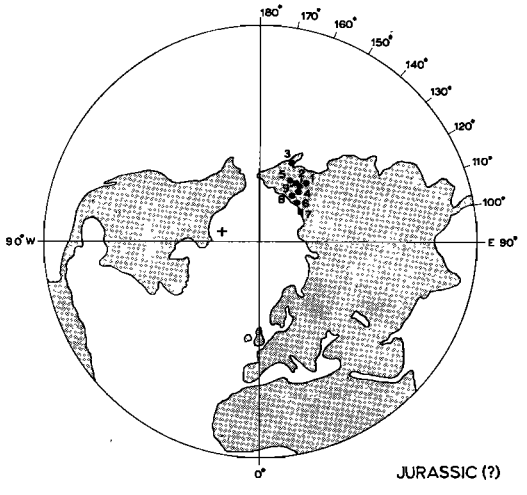


Fig. 3b.

Figure 3 shows distribution of Triassic, Jurassic(?), and Cretaceous magnetic poles for the eastern United States. Short dashed line: great circle through mean pole and present magnetic pole.

sion. The remanent magnetization can thus be assumed to have been parallel to the earth's magnetic field at the time that the intrusions cooled.

The inclinations of the magnetic directions in the dikes range from 10° to 60° and are predominantly positive. Only four cores show negative inclinations; they remained unchanged during demagnetization. These four cores were taken from the chilled borders of dikes. Cores from the central parts of the same intrusions showed normal positive inclinations. It is possible that a self-reversing phenomenon is involved.

The declinations and inclinations of the regional magnetic directions of dike swarms sampled in widely separated regions show a remarkable consistency. The range in average declinations is about 20°; the range in average inclinations is about 15°. This consistency is reflected in a linear alignment of the magnetic poles (see Figure 3). The line along which the magnetic poles are located is part of a great circle with its normal orientated 15°N, 96°W. A similar linear relationship, which is also more or less latitudinal, has been found for the Jurassic magnetic poles of Europe [*Krs*, 1966]. The line along which these Jurassic poles are concentrated is part of a great circle with its normal orientated 30°N, 130°W. Such align-

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ments of magnetic poles can be attributed to a dominantly linear secular variation, but they can also be due to megatectonic movements of the continents involved. The dikes examined in the present study do not appear to be related to such megatectonic activity; therefore such movements will not be discussed here.

The value K is a precision parameter that gives an indication of dispersion in directions of magnetization. Low values of K indicate high scatter, and, as K becomes larger, the grouping improves. K varies from about 30 to 100 for the magnetic directions measured in the dikes. The value for α_{95} is the radius of the circle of confidence at the 95% confidence level. For the paleomagnetic directions of the dikes these values range from 6° to 19°.

From this paleomagnetic study the following results can be deduced:

1. The relatively close grouping of the regional magnetic poles for dike swarms sampled throughout the Appalachians suggests a common origin for these intrusions. The linear arrangement of the poles may indicate slight differences in age between the separate dike swarms. There is, however, little doubt that most dikes belong to a single period of igneous activity (see Figure 3).

2. The dikes intruded neither in Late Triassic nor in Early Cretaceous time. Their paleomagnetic directions do not coincide with those obtained from known Late Triassic and Early Cretaceous intrusions. They should accordingly be attributed to the Jurassic (see Figure 3).

3. The close grouping of the paleomagnetic poles suggests, furthermore, that no post-emplacement deformations of the dikes took place and that the dikes have not been significantly tilted and/or rotated. Their fan-shaped arrangement therefore is primary and is not due to differential rotation.

Structural Interpretation

Lester and Allen [1950] related the Piedmont dike intrusions to the pre-Cretaceous downwarping of the eastern seaboard. They explain the strike of the dikes, more or less at right angles to the hingeline of this downwarp, as follows:

> It is postulated that the areas of dike concentration were the axis of NW-SE trending synclines, which existed simultaneously at the time of the diabase intrusion. A force couple, acting along the southern edge of the Piedmont, could develop folds which would plunge to the northwest and have longitudinal axis almost normal to the Fall line. This condition would be similar to that obtained by compression of a piece of cloth cut on a bias which would essentially parallel the trend of the Fall line.

It is difficult to accept this explanation for regional swarms that show gradual changes in dike trends, especially since Lester and Allen do not specify the sort of force couple involved.

According to *King* [1961] the dikes reflect deep-seated tensile stresses, and the mafic magma intruded second-order tensional fractures which had originated by an elastic release at right angles to the original direction of principal horizontal stress (*PHS*). In the northeastern and southwestern extremities of the Appalachians the longitudinal crustal arching of Late Triassic time closely followed an axis directed NE–SW. In the central segment and in New England the arching appears to have followed the pre-existing structural sinuosities of the crystalline basement. A release of the *PHS* involved in this uplift resulted chiefly in NE-trending fractures, as evidenced by the boundary faults bordering the Late Triassic grabens. These fractures formed the pathway for the ascending mafic magmas that intruded and outpoured in Late Triassic time.

The fractures followed by the dike swarms have no apparent structural relationship to this Triassic pattern. Since paleomagnetic directions indicate that the dikes are not related to the Late Triassic magmatic activity, it is more probable that the fractures originated in another distinct tectogenesis, when the tectonic activity may well have been compressional. In this case, the dikes could have intruded shear faults as well as tension faults.

To differentiate between dikes that intruded shear faults and dikes that intruded tension faults is not easy. However, dikes following tension faults tend to be more continuous laterally because such a fracture opens up by a gradual wall separation at right angles to the fault plane. Because tensional fractures commonly follow zones of least resistance, it can be expected that these fracture planes will be somewhat sinuous and that the emplaced dikes will

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show many small-amplitude curves. Magmas intruding shear faults, on the other hand, will form dikes that are laterally more discontinuous because irregularities in the fault plane will locally seal off the fracture. Also, shear faults in general are remarkably straight.

Unfortunately, parallel dikes of the same swarm often display structural differences that point to intrusion along both fault types. Detailed knowledge of the distribution of the dikes in the various regions can be more decisive. A series of strike-frequency diagrams shows a consistent NNW dike trend in the Piedmont (Alabama, Georgia, South and North Carolina) and a NNE dike trend in New England. In Virginia and Pennsylvania the dike trend distribution shows three maxima (see Table 3). The considerable differences in apparent conjugate shear angles in these distributions imply that some of the sets follow older fractures. If only the dike concentrations that cut the youngest deposits (the Newark formation in the Triassic grabens) are considered, it is the central set that dominates frequency diagrams of the central Appalachians. Since these dikes intrude uniform, well-bedded formations of considerable thickness and since their directions do not change when they penetrate the surrounding metamorphic complex, there is little doubt that the magma intruded tensional fractures. The fact that the dike swarms in the Piedmont and New England are consistently found in one set substantiates this view.

According to *King* [1961] the dikes intruded second-order tensional faults. King's preference for tensile stresses at the time of emplacement obviously rises from general views on the Late Triassic relaxation of compressive stress in the Appalachians. The fractures followed by the dikes do not show any structural relationship to the Triassic structural pattern and are distinctly younger. Furthermore, fossil magnetic directions in the basalt suggest that the dike intrusion took place during a time of volcanic and tectonic activity unrelated to the Triassic tectogenesis. Therefore, two periods of relaxation of tectonic stress are required to explain both the Triassic tectonic pattern and the dike pattern. Even without discussing the possibility of a Late Triassic release of compressive stress, it seems superfluous to assume that there have been several such periods during the Mesozoic. It is my opinion that the dike swarms intruded tensional fractures of primary origin.

It is important to know whether these first-order tensional fractures, into which the dikes intruded, originated in a pure-nonrotational or in a simple-rotational shear system. In order to investigate the origin of the fractures the author conducted a series of clay-cake experiments.

In a pure-shear system the first-order tensional fractures develop parallel to the *PHS*. As evidenced by the clay-cake experiments, the angle between these tensional faults and the *PHS* is generally small, rarely exceeding 10°. Except for local anomalous divergences, the intrusions following these faults should form extensive swarms of amazingly parallel dikes.

In a simple-shear system the primary tension

TABLE 3. Main Directions of the Mesozoic Dikes in the Appalachians

Region	N*	Dike Trend†	Regional Trend	References
Georgia Piedmont	106	326°	(45°)	*Lester and Allen* [1950]
Kings Mountains	49	327°	(45°)	*Keith* [1931]
Deep River basin (W)	118	330°	(45°)	*Reinemund* [1955]
Deep River basin (central)	210	333°	(45°)	*Reinemund* [1955]
Deep River basin (E)	160	339°	(45°)	*Reinemund* [1955]
Virginia (NW)	139	310°–347°–39°	(30°)	*Calver et al.* [1963]
Virginia (SE)	139	343°–358°–18°	(35°)	*Calver et al.* [1963]
Pennsylvania (SE)	166	12°– 35°–70°	(65°)	*Gray* [1960]
Connecticut and Rhode Island	78	30°	(10°)	*Rodgers et al.* [1959]

* Number of measurements.
† Maximum azimuth of dike trend.

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faults develop at about 45° to the direction of the main shear plane. During deformation the fractures rotate, conforming to the sense of rotation of the system, and thus increase the angle between the tensional fractures and the main shear plane from about 45° to about 65°. When that value is reached, new tensional fractures develop at the original angle with the main shear plane. With decreasing values of the compressional component of the shear system the magma will tend more and more to intrude the fractures that develop at 10° to 30° to the main shear plane. These subvertical fractures originate as synthetic shear faults and have a limited strike-slip component. With increasing deformation, however, the movement along the strike ceases, and the faults show a tendency to become tensional. They open up at right angles to the fracture plane and would form ready pathways for ascending magmas.

In the Appalachians the dike swarms are restricted to a zone trending NE-SW, following the main mountain axis. A pure-shear deformation of such a zone by a stress directed from SE to NW can be expected to form primary tension fractures trending mostly NW–SE. If simple-shear deformation took place in a zone trending NE–SW, tension fractures can be expected to cut that zone diagonally. These fractures originate at an angle of about 45° to the main shear plane (here supposedly trending more or less NE–SW); they will rotate synchronously with the polarity of the movements. If the rotational movements develop with a sinistral polarity, tension faults trending from about NNE to NNW can be expected; if the rotational movement develops with a dextral polarity, the trends would sway from ENE to WNW. The fan-shaped distribution of the Appalachian dike swarms does not fit either of these patterns. As in most mountain belts, the fracture system occupied by the dikes was most probably formed by a combination of pure- and simple-shear deformation.

A possible explanation for the fan-shaped dike distribution may lie in a gradual change of the shear system in the NE–SW-trending crustal segment occupied by the Appalachians —a change from a pure-shear system in the southwest to a simple-shear system in the northeast (see Figure 4). This requires a decrease of the compressional component of the *PHS* (the NW–SE component), and an increase of the rotational component of the *PHS* (the NE–SW component) in northeast direction. If we assume that the dike trends are indicative of the direction of *PHS* (in other words, that the dikes intruded first-order tension fractures), then the compressional movements dominated in the southwestern Appalachian segment and the rotational movements were more significant in the northeastern segment of the Appalachians. In this case we are dealing with a simple-shear system with sinistral polarity.

Under conditions of predominant simple shear in the northeastern segment of the Appalachians, rotations of large-scale crustal units can be expected. Such rotations have actually been measured for Newfoundland. *Black* [1964] compared the late Paleozoic magnetic directions of rocks sampled there with those from New Brunswick and Quebec. Its deviating paleomagnetic poles as well as its varying structural trend indicate that Newfoundland has rotated counterclockwise about 30° with respect to the mainland, assuming the prevalence of simple-shear movements with sinistral polarity at the time of rotation. Structural considerations have led *Wilson* [1962*a*, *b*] to propose that a NE–SW-trending transcurrent fault with sinistral polarity, the Cabot fault, extends from northern Newfoundland to Massachusetts. This fault would have been active in late Paleozoic time. These data, although neither conclusive nor complete, do suggest that strike-slip movements with sinistral polarity were dominant in the northeastern Appalachians during late Paleozoic and possibly during Mesozoic time.

So far, it is possible to distinguish at least three episodes of tectonic and magmatic activity in the Appalachians during the Mesozoic. The most important and most easily recognizable is undoubtedly that in the Late Triassic. The second episode, probably Jurassic, is represented by the intrusion of the dike swarms, and a third, more local, episode is indicated by the Early Cretaceous emplacement of mafic intrusions (Monteregian Hills, Mount Ascutney, and Mount Megantic). Such times of increased tectonic and subsequent mafic magmatic activity appear indicative of a convulsive release of compressive stress in Mesozoic time.

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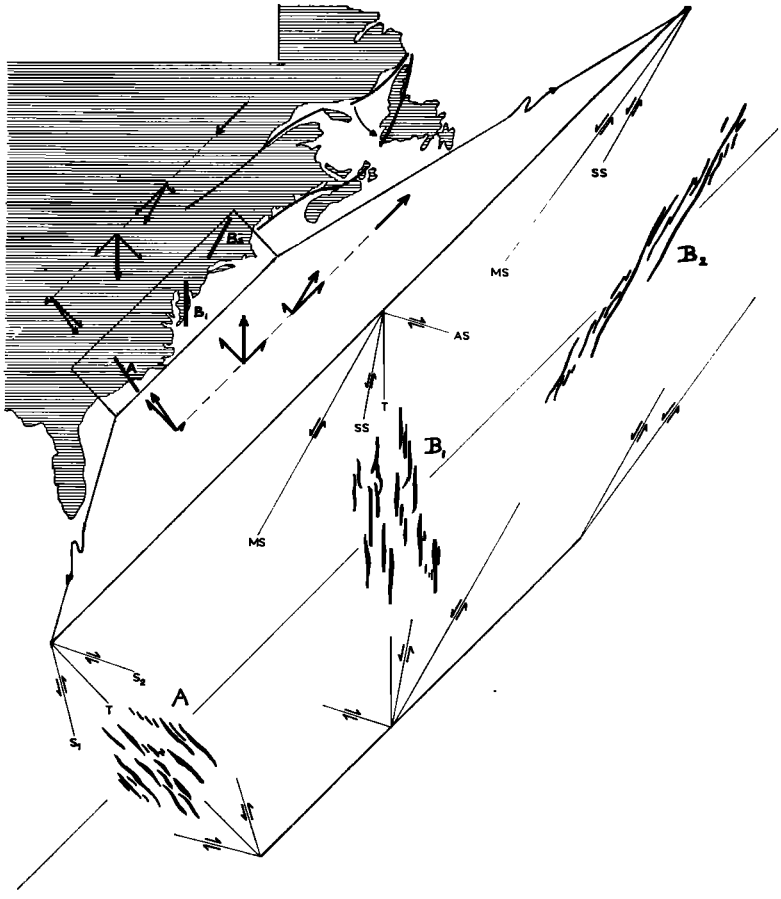


Fig. 4. The structural relationship of the Mesozoic dikes swarms to the regional deformation pattern in the Appalachians.

A: Pure shear system—rhombic strain. *PHS* $\angle$ S_1, S_2 = 15°–45°; *PHS* $\angle$ T = 0°–10°. *PHS* is principal horizontal stress, S_1 and S_2 are shear fractures in pure shear system, and T is tensional fracture.

B: Simple shear system—monoclinic strain (sinistral polarity). *MS* $\angle$ *SS* = 10°–30°; *MS* $\angle$ T = 25°–65°; *MS* $\angle$ *AS* = 60°–100°. *MS*, main shear fracture; *SS*, synthetic shear fracture; and *AS*, antithetic shear fracture.

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(Received May 21, 1966;
revised October 12, 1966.)

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